School of Engineering

E211 – Operations Planning II

Lesson 4 Worksheet

<u>Warm up question (pick 1)</u>

Below are some real-world queuing systems. State the arrival, queue and service characteristics in each queuing system.

a. Food ordering

<u>Arrival:</u>
Population size: Infinite
Arrival pattern: Random
Behaviour: Depends on the queue (Mostly customers are patient, some might be impatient)

<u>Queue:</u>
Queue capacity: Unlimited
Queue configuration: Multiple line due to multiple counters
Queue discipline: First-Come-First-Serve

<u>Service:</u>
Service time pattern: Random, Exponential
Number of servers: Multiple
Number of phases: 2 (After order placement → collect food from counter)

b. Car refuelling at petrol station

c. Security check at airport

d. Queuing at a theme park ride.

School of Engineering

School of Engineering

Problem Statement

Figure 1 shows the self-checkout area at an IKEA store.

Part 1

Figure 1: IKEA self-checkout area

Data gathered:

- Arrival rate = 150 customers per hour
- Mean checkout time = 2.5 minutes

a. Which queuing model should you use to study the IKEA self-checkout area? State the assumptions.
M/M/K (M/M/8).

Assumptions:
Input source: Infinite population; no balks, no reneging
Arrival process: Number of arrivals follows Poisson distribution
Queue: Unlimited; single line
Queue discipline: FIFO (FCFS)
Service mechanism: Multiple servers, single phase system
Service time: Negative exponential distribution
Arrivals and service times are independent of the number of customers in system
Arrival rate must be smaller than sum of service rate of servers, i.e. λ/kμ

b. Determine the minimum number of checkout machines needed in order for the queuing system to be in steady state.

$\lambda < K\mu$ **(Arrival rate < No of servers*Service Rate)**

Arrival rate (λ) = 150/60 = 2.5/min
Service rate (μ) = 1 / 2.5 = 0.4/min
2.5 < k(0.4)
2.5 < 0.4k
k = 2.5/0.4 = 6.25 = 7 (minimum number)

c. Assume there are 8 checkout machines. Determine the performance measures of the queuing system – utilization, waiting time, time in system, queue length and number in system.

Utilization **(ρ)** = 0.7813
Waiting time **(Wq)** = 0.60
Time in system **(Ws)** = 3.10
Queue length **(Lq)** = 1.49
Number in system **(Ls)** = 7.74

Problem Statement:

Arrival rate = 150 customers per hour
Mean checkout time = 2.5 minutes

M/M/k

Arrival rate	λ	2.500	per minute
Service rate	μ	0.4	per minute
No. of servers	k	8	
Utilisation	ρ	0.7813	
Prob (no customers)	P_0	1.58E-03	
No. of customers in queue	Lq	1.49	
No. of customers in system	Ls	7.74	
Waiting time in queue	Wq	0.60	minutes
Time in system	Ws	3.10	minutes

d. Calculate the probability that at least 1 self-checkout machine is available at any time.

No of servers (k) = 8

Probability = P0 + P1 + P2 + P3 + P4 + P5 + P6 + P7
= 0.581781247

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.00158428	0.659221436
1	6.25	0.00990175	0.515016747
2	19.53125	0.030942969	0.402356834
3	40.69010417	0.06446452	0.314341276
4	63.57828776	0.100725812	0.245579122
5	79.4728597	0.125907265	0.191858689
6	82.78422885	0.131153401	0.149889601
7	73.91449005	0.117101251	0.117101251

e. Calculate the probability that there are at least 2 customers in the queue.
No of servers (k) = 8

Probability = 1 – (P0 + P1 + P2 + P3 + P4 + P5 + P6 + P7 + P8 + P9)
= 0.255260469

Since n (9) > k (8), we take the value of **0.071472931** when calculating.

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.00158428	0.659221436
1	6.25	0.00990175	0.515016747
2	19.53125	0.030942969	0.402356834
3	40.69010417	0.06446452	0.314341276
4	63.57828776	0.100725812	0.245579122
5	79.4728597	0.125907265	0.191858689
6	82.78422885	0.131153401	0.149889601
7	73.91449005	0.117101251	0.117101251
8	57.74569535	0.091485352	0.091485352
9	40.10117733	0.063531495	0.071472931

<u>Part 2</u>

Suppose that there is a cashier counter set up to assist customers who are unable to perform self-checkout. Your task is to determine an average service time standard for the cashier checkout so that the service level and staff utilization are reasonable. Initial estimates are that on average, 20 customers are expected to use the cashier checkout every hour.

Discuss the approach you would take to carry out your task.
I will use M/M/1.

Arrival rate = 20/60 = 0.333/min

Assume system utilization = 0.8,
System utilization = Arrival rate/Service rate
System utilization = 0.333/service rate
0.8 = 0.333 / service rate
Service rate = [(20/60) / 0.8)] = 0.41666

Wq (average time in queue) = 9.601, Ws (average time in system) = 12.001
Add more counters to reduce waiting times.